

Positivity of h_m for $3 \mid d$, and an Orbit–Product Formula for h_m when $\gcd(d, 3) = 1$

Seed 2, Layer 2, Round 2

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Abstract

Let $c = (c_0, c_1, c_2)$ be a profile with $d = c_0 + c_1 + c_2$, let $F_{c,m}(q)$ be the generating function of cylindric partitions of profile c with maximum entry at most m , let $g_m = F_{c,m} - F_{c,m-1}$ and $h_m = (q^\ell; q^\ell)_m g_m$ with $\ell = \gcd(d, 3)$. We prove two results. (1) For $3 \mid d$, h_m has non-negative coefficients; this settles the $\ell = 3$ half of the core bottleneck identified in the Layer-1 synthesis. (2) For $\gcd(d, 3) = 1$ we prove an exact *orbit–product formula*: h_m (and also $P_m := (q; q)_m F_{c,m}$) is a finite sum, over sequences of C_3 -orbits of profiles, of products $\prod_{j=1}^m U^{(j)}(q^j)$ of explicit integer polynomials $U^{(j)}$ with coefficients in $\{0, \pm 1\}$. In particular $(q; q)_m F_{c,m} \in \mathbb{Z}[q]$, and $h_m(1) = K^m$ with $K = \binom{d+2}{2}/3$. The remaining obstacle to $h_m \geq 0$ for $\gcd(d, 3) = 1$ is isolated as the positivity of this signed orbit-path sum. Both results take as input only the Adjugate Monomial Theorem $\text{adj}(I - A(x))[c, c'] = x^{\text{EMD}(c, c')}$ and $\det(I - A(x)) = -(x^3 - 1)$, which were proved in Layer 1 (Seed 4).

1 Setup and inputs

Profiles of weight d are triples $c = (c_0, c_1, c_2) \in \mathbb{Z}_{\geq 0}^3$ with $\sum c_i = d$. The Corteel–Welsh recursion for the bounded generating functions $F_{c,m}$ is encoded by the matrix $A(x)$ with entries indexed by profiles, and Layer 1 (Seed 4) proved:

Input 1 (Adjugate Monomial Theorem, GREEN). $\text{adj}(I - A(x))[c, c'] = x^{\text{EMD}(c, c')}$, where

$$\text{EMD}(c, c') = 3 \max(0, c'_1 - c_1, c_0 - c'_0) + (c'_0 - c_0) - (c'_1 - c_1), \quad (1)$$

and $\det(I - A(x)) = -(x^3 - 1)$. Consequently, for every $m \geq 1$,

$$(1 - q^{3m}) F_{c,m} = \sum_{c'} q^{m \text{EMD}(c, c')} F_{c',m-1}, \quad F_{c,0} = 1. \quad (2)$$

EMD is the minimum-cost clockwise transport distance on $\mathbb{Z}/3$: writing $f = (f_0, f_1, f_2)$ for a clockwise flow (f_i units crossing edge $i \rightarrow i+1$), the transport equations are $c_i - c'_i = f_i - f_{i-1}$, whose solutions form the line $f = f^* + t(1, 1, 1)$; then $\text{EMD}(c, c') = \min\{f_0 + f_1 + f_2 : f \geq 0\}$, and (??) is the closed form. Note $\text{EMD}(c, c) = 0$.

2 Symmetry lemmas

Let σ denote the cyclic rotation $\sigma(c_0, c_1, c_2) = (c_2, c_0, c_1)$.

Lemma 1 (Profile-rotation invariance of F). $F_{\sigma c, m} = F_{c, m}$ for all c and all $m \geq 0$ (as power series in q).

Proof. A cylindric partition of profile $(c_1, \dots, c_k)$ is a tuple $(\lambda^{(1)}, \dots, \lambda^{(k)})$ with $\lambda_j^{(i)} \geq \lambda_{j+c_{i+1}}^{(i+1)}$ ($1 \leq i \leq k-1$) and $\lambda_j^{(k)} \geq \lambda_{j+c_1}^{(1)}$. The relabelling $\Lambda = (\lambda^{(1)}, \dots, \lambda^{(k)}) \mapsto \Lambda' = (\lambda^{(2)}, \dots, \lambda^{(k)}, \lambda^{(1)})$ is a bijection from cylindric partitions of profile $(c_1, \dots, c_k)$ to those of profile $(c_2, \dots, c_k, c_1)$: each defining inequality of Λ' is one of the defining inequalities of Λ (the interior ones shift the index by one; the two boundary inequalities swap roles). The bijection preserves total size and maximum entry, hence preserves $F_{\cdot, m}$. Iterating gives invariance under the full rotation group C_3 , in particular under σ . $\square$

Lemma 2 (Rotation invariance of EMD). $\text{EMD}(\sigma c, \sigma c') = \text{EMD}(c, c')$.

Proof. Let $a = c'_0 - c_0$, $b = c'_1 - c_1$, so $\text{EMD}(c, c') = 3 \max(0, b, -a) + a - b$ by (??). For the rotated pair, $(\sigma c')_0 - (\sigma c)_0 = c'_2 - c_2 = -(a+b)$ (since the coordinate differences sum to 0) and $(\sigma c')_1 - (\sigma c)_1 = c'_0 - c_0 = a$. Hence

$$\text{EMD}(\sigma c, \sigma c') = 3 \max(0, a, a+b) - (a+b) - a = 3(a + \max(-a, 0, b)) - 2a - b = 3 \max(0, b, -a) + a - b,$$

which is $\text{EMD}(c, c')$. $\square$

Lemma 3 (Mod-3 shift). $\text{EMD}(c, \sigma c') \equiv \text{EMD}(c, c') + d \pmod{3}$. Consequently, if $\gcd(d, 3) = 1$ the three values $\text{EMD}(c, c')$, $\text{EMD}(c, \sigma c')$, $\text{EMD}(c, \sigma^2 c')$ are pairwise distinct mod 3 and represent all residues $\{0, 1, 2\}$; if $3 \mid d$ they are all congruent mod 3.

Proof. Modulo 3, (??) gives $\text{EMD}(c, c') \equiv (c'_0 - c_0) - (c'_1 - c_1)$. Hence

$$\text{EMD}(c, \sigma c') - \text{EMD}(c, c') \equiv [(c'_2 - c_0) - (c'_0 - c_1)] - [(c'_0 - c_0) - (c'_1 - c_1)] = c'_1 + c'_2 - 2c'_0 \equiv d - 3c'_0 \equiv d \pmod{3} \quad (\text{m})$$

$\square$

(Lemmas ?? and ?? were additionally verified numerically for all pairs of profiles with $d \leq 20$.)

3 Theorem 1: $h_m \geq 0$ for $3 \mid d$

Define, for any d , $\tilde{P}_m(c) := (q^3; q^3)_m F_{c,m}$.

Lemma 4. *For all d and all $m \geq 0$, $\tilde{P}_m(c)$ has nonnegative coefficients; indeed*

$$\tilde{P}_m(c) = \sum_{c'} q^{m \text{EMD}(c,c')} \tilde{P}_{m-1}(c'), \quad \tilde{P}_0 = 1. \quad (3)$$

Proof. Multiply (??) by $(q^3; q^3)_{m-1}$ and use $(q^3; q^3)_m = (1 - q^{3m})(q^3; q^3)_{m-1}$. Nonnegativity follows by induction on m , since (??) has nonnegative coefficients on the right-hand side. (Unfolding (??) recovers the manifestly positive EMD path formula of Round 1, Agent B.) $\square$

Theorem 1. *Let $3 \mid d$, so $\ell = \gcd(d, 3) = 3$ and $h_m = (q^3; q^3)_m (F_{c,m} - F_{c,m-1})$. Then for all profiles c and all $m \geq 1$, h_m has nonnegative coefficients. Explicitly,*

$$h_m(c) = \sum_{c' \neq c} q^{m \text{EMD}(c,c')} \tilde{P}_{m-1}(c') + q^{3m} \tilde{P}_{m-1}(c). \quad (4)$$

Proof. By definition and Lemma ??,

$$h_m = (q^3; q^3)_m F_{c,m} - (q^3; q^3)_m F_{c,m-1} = \tilde{P}_m(c) - (1 - q^{3m}) \tilde{P}_{m-1}(c).$$

Substitute (??) for $\tilde{P}_m(c)$; the $c' = c$ term equals $q^{m \text{EMD}(c,c)} \tilde{P}_{m-1}(c) = \tilde{P}_{m-1}(c)$, and $\tilde{P}_{m-1}(c) - (1 - q^{3m}) \tilde{P}_{m-1}(c) = q^{3m} \tilde{P}_{m-1}(c)$, giving (??). Every term on the right of (??) is a nonnegative series by Lemma ??. $\square$

Remark 1. *Theorem ?? upgrades the $3 \mid d$ half of the Layer-1 core bottleneck (“ $(q^\ell; q^\ell)_m g_m \geq 0$ for all d ”) from YELLOW to GREEN. It is a strictly one-line consequence of the Adjugate Monomial Theorem; the difficulty of the conjectural case $\gcd(d, 3) = 1$ is therefore *entirely* concentrated in the division by $\prod_{j=1}^m (1 + q^j + q^{2j}) = (q^3; q^3)_m / (q; q)_m$.*

4 Theorem 2: the orbit–product formula for $\gcd(d, 3) = 1$

Assume now $\gcd(d, 3) = 1$ and set $P_m(c) := (q; q)_m F_{c,m}$, so $h_m = P_m - (1 - q^m) P_{m-1}$. Since $d \not\equiv 0 \pmod{3}$, no profile satisfies $\sigma c = c$, so σ acts freely and the $\binom{d+2}{2}$ profiles fall into $K := \binom{d+2}{2}/3$ orbits $O = \{c', \sigma c', \sigma^2 c'\}$.

Lemma 5 (Local divisibility). *For profiles c, c' define $T_{c,[c']}(x) := \sum_{r=0}^2 x^{\text{EMD}(c, \sigma^r c')}$, and the top-modified version $T_{c,[c']}^{\text{top}}(x)$, which is defined as T except that when $c \in [c']$ the term $x^{\text{EMD}(c,c)} = x^0$ is replaced by x^3 . Then T and T^{top} depend only on c and the orbit $[c']$, and both are divisible by $1+x+x^2$ in $\mathbb{Z}[x]$. We write $U_{c,[c']} := T_{c,[c']}/(1+x+x^2)$ and $U_{c,[c']}^{\text{top}} := T_{c,[c']}^{\text{top}}/(1+x+x^2)$.*

Proof. Independence of the representative c' is clear ($\sigma^r c'$ runs over the orbit). By Lemma ?? the three exponents represent all residues mod 3. Since $x^k \equiv x^{k \bmod 3} \pmod{1+x+x^2}$ and $1+x+x^2 \equiv 0$, any sum of three monomials whose exponents cover all residues mod 3 is divisible by $1+x+x^2$. The top modification replaces the exponent 0 by 3, preserving the residue, so divisibility persists. (Independence of representative in the top case: the modified term is attached to the unique representative equal to c itself.) $\square$

Theorem 2 (Orbit-product formula). *Let $\gcd(d, 3) = 1$ and let $\mathcal{O}$ be the set of σ -orbits of profiles of weight d . Then for all $m \geq 0$ and all profiles c ,*

$$P_m(c) = \sum_{(O_{m-1}, \dots, O_0) \in \mathcal{O}^m} \prod_{j=1}^m U_{o_j, O_{j-1}}(q^j), \quad (5)$$

where $o_m := c$ and o_j denotes any representative of O_j for $j < m$ (the product is well defined by Lemma ?? and Lemma ??); and

$$h_m(c) = \sum_{(O_{m-1}, \dots, O_0) \in \mathcal{O}^m} U_{c, O_{m-1}}^{\text{top}}(q^m) \prod_{j=1}^{m-1} U_{o_j, O_{j-1}}(q^j). \quad (6)$$

Proof. First, one level. By Lemma ??, $P_{m-1}(\sigma c') = P_{m-1}(c')$, so grouping the right side of (??) (multiplied by $(q; q)_{m-1}$) into orbits,

$$(1 - q^{3m}) (q; q)_{m-1} F_{c,m} = \sum_{O \in \mathcal{O}} T_{c,O}(q^m) P_{m-1}(O).$$

The left side equals $(1 + q^m + q^{2m}) P_m(c)$, and by Lemma ?? $T_{c,O}(q^m) = (1 + q^m + q^{2m}) U_{c,O}(q^m)$. Since $\mathbb{Z}[[q]]$ is an integral domain and $1 + q^m + q^{2m} \neq 0$, we may cancel:

$$P_m(c) = \sum_{O \in \mathcal{O}} U_{c,O}(q^m) P_{m-1}(O). \quad (7)$$

Well-definedness in O of the coefficient uses Lemma ??: replacing the representative of an intermediate orbit rotates both arguments of the relevant

EMD's simultaneously and changes nothing. Iterating (??) down to $P_0 = 1$ gives (??).

For (??): $h_m = P_m(c) - (1 - q^m)P_{m-1}(c)$. Multiply by $1 + q^m + q^{2m}$:

$$(1 + q^m + q^{2m}) h_m = \sum_O T_{c,O}(q^m) P_{m-1}(O) - (1 - q^{3m}) P_{m-1}(c).$$

In the orbit O_c of c , the term $q^{\text{EMD}(c,c)m} P_{m-1}(c) = P_{m-1}(c)$ combines with $-(1 - q^{3m}) P_{m-1}(c)$ to give $q^{3m} P_{m-1}(c)$: this is exactly the replacement $x^0 \mapsto x^3$ defining T^{top} . Hence

$$(1 + q^m + q^{2m}) h_m = \sum_O T_{c,O}^{\text{top}}(q^m) P_{m-1}(O),$$

and cancelling $1 + q^m + q^{2m}$ (Lemma ??) and expanding P_{m-1} by (??) gives (??). $\square$

Corollary 1. *For $\gcd(d, 3) = 1$: $P_m = (q; q)_m F_{c,m}$ and h_m lie in $\mathbb{Z}[q]$ (they are honest polynomials), and $P_m(1) = h_m(1) = K^m$ where $K = \binom{d+2}{2}/3 = (d+1)(d+2)/6$.*

Proof. Each U, U^{top} is an integer polynomial and the sums in Theorem ?? are finite. At $q = 1$: $T_{c,O}(1) = 3$ and $T_{c,O}^{\text{top}}(1) = 3$, so every $U(1) = U^{\text{top}}(1) = 1$, and each of the K^m orbit sequences contributes exactly 1. $\square$

Remark 2 (Computed structure of the U 's). *Numerically (all $d \leq 14$ with $\gcd(d, 3) = 1$, all pairs (c, O) , both plain and top): every $U_{c,O}$ has coefficients in $\{0, \pm 1\}$, and its nonzero coefficients alternate in sign, beginning and ending with $+1$ (so $U = x^{a_0} - x^{a_1} + \dots + x^{a_{2t}}$ with $a_0 < a_1 < \dots < a_{2t}$ and $U(1) = 1$). Also verified: (??) and (??) reproduce the transfer-matrix values of P_m and h_m exactly for $d \in \{1, 2, 4, 5\}$, all orbits, $m \leq 3$ ($m \leq 2$ for $d = 5$), and the resulting h -values agree with Seed 8's independently computed tables (e.g. $d = 4, c = (2, 1, 1)$: $h_2 = [0, 0, 3, 4, 5, 3, 3, 2, 2, 1, 1, 0, 1]$).*

Remark 3 (What remains). *The single remaining obstacle to the core bottleneck ($h_m \geq 0$, hence $Q_n \geq 0$ via the D_k^m tower) is the positivity of the signed sum (??). Three natural strengthenings FAIL and should not be retried naïvely: (i) individual orbit-sequence products $\prod_j U(q^j)$ can have negative coefficients; (ii) Abel summation along a domination chain fails because the orbit values $P_{m-1}(O)$ are pairwise incomparable coefficientwise for $m \geq 2$; (iii) shifted domination $q^\delta P(O) \leq P(O')$ fails for all δ (the P 's are polynomials of comparable degree). What is promising: the U 's are $\{0, \pm 1\}$*

alternating with $U(1) = 1$, evaluated at distinct scales q^j — the sum (??) is a signed lattice-path model begging for a sign-reversing involution that fixes exactly K^m monomials (one per orbit sequence).